



Department of Computer Technology

Vision of the Department

To be a well-known centre for pursuing computer education through innovative pedagogy, value-based education and industry collaboration.

Mission of the Department

To establish learning ambience for ushering in computer engineering professionals in core and multidisciplinary area by developing Problem-solving skills through emerging technologies.

Session 2025-2026

Vision: Dream of where you want.	Mission: Means to achieve Vision
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Program Educational Objectives of the program (PEO): (broad statements that describe the professional and career accomplishments)

PEO1	Preparation	P: Preparation	Pep-CL abbreviation pronounce as Pep-si-IL easy to recall
PEO2	Core Competence	E: Environment (Learning Environment)	
PEO3	Breadth	P: Professionalism	
PEO4	Professionalism	C: Core Competence	
PEO5	Learning Environment	L: Breadth (Learning in diverse areas)	

Program Outcomes (PO): (statements that describe what a student should be able to do and know by the end of a program)

Keywords of POs:

Engineering knowledge, Problem analysis, Design/development of solutions, Conduct Investigations of Complex Problems, Engineering Tool Usage, The Engineer and The World, Ethics, Individual and Collaborative Team work, Communication, Project Management and Finance, Life-Long Learning

PSO Keywords: Cutting edge technologies, Research

“I am an engineer, and I know how to apply engineering knowledge to investigate, analyse and design solutions to complex problems using tools for entire world following all ethics in a collaborative way with proper management skills throughout my life.” to contribute to the development of cutting-edge technologies and Research.

Integrity: I will adhere to the Laboratory Code of Conduct and ethics in its entirety.

Name and Signature of Student and Date

(Signature and Date in Handwritten)



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Session	2025-26 (ODD)	Course Name	HPC Lab
Semester	7	Course Code	22ADS706
Roll No	03	Name of Student	Debasrita Chattopadhyay

Practical Number	05
Course Outcome	1. Understand and Apply Parallel Programming Concepts 2. Analyze and Improve Program Performance. 3. Demonstrate Practical Skills in HPC Tools and Environments.
Aim	Basics of MPI Programming
Problem Definition	MPI program demonstrates how to initialize the MPI environment, send and receive messages, and finalize the MPI environment.
Theory (100 words)	MPI (Message Passing Interface) is a standard library for parallel programming in distributed memory systems. Multiple processes (running the same or different machines) communicate through message passing. MPI fits in many of the high-performance computing (HPC) clusters that need large-scale parallelism. Note: - Processes will communicate explicitly using messages. - There is point-to-point as well as collectives.
Procedure and Execution (100 Words)	Algorithm: 1. Download OpenMPI wget https://download.open-mpi.org/release/open-mpi/v5.0/openmpi-5.0.5.tar.gz 2.Extract the Source Code tar -xvzf openmpi-5.0.5.tar.gz cd openmpi-5.0.5 3. Configure for Local Installation Tell OpenMPI to install inside your home folder: ./configure --prefix=\$HOME/.openmpi 4. Compile OpenMPI



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	This will take a few minutes depending on your machine: make -j4 5. Install Locally make install 6. Update Your Environment Add the following lines to the end of your ~/.bashrc file: export PATH=\$HOME/.openmpi/bin:\$PATH export LD_LIBRARY_PATH=\$HOME/.openmpi/lib:\$LD_LIBRARY_PATH 7. Then reload: source ~/.bashrc 8. Verify Installation mpicc --version mpirun --version
	Code: MPI Programs Contains 4 MPI examples: 1. Hello World 2. Broadcast 3. Scatter & Gather 4. Reduce Write a Basic MPI Program (Hello World) <pre>#include <stdio.h> #include <mpi.h> int main(int argc, char* argv[]) { int rank, size; MPI_Init(&argc, &argv); // Initialize MPI</pre>



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```
MPI_Comm_rank(MPI_COMM_WORLD, &rank); // Get
process rank
MPI_Comm_size(MPI_COMM_WORLD, &size); // Get total
processes

printf("Hello from process %d of %d\n", rank, size);

MPI_Finalize(); // Finalize MPI
return 0;
}
Point-to-Point Communication (Send/Recv)
#include <stdio.h>
#include <mpi.h>

int main(int argc, char* argv[]) {
    int rank, size, data;

    MPI_Init(&argc, &argv);
    MPI_Comm_rank(MPI_COMM_WORLD, &rank);
    MPI_Comm_size(MPI_COMM_WORLD, &size);

    if (rank == 0) {
        data = 100;
        MPI_Send(&data, 1, MPI_INT, 1, 0,
MPI_COMM_WORLD);
        printf("Process 0 sent data %d to process 1\n", data);
    } else if (rank == 1) {
        MPI_Recv(&data, 1, MPI_INT, 0, 0,
MPI_COMM_WORLD, MPI_STATUS_IGNORE);
        printf("Process 1 received data %d from process 0\n", data);
    }

    MPI_Finalize();
    return 0;
}

Collective Communication (Broadcast)
#include <stdio.h>
#include <mpi.h>

int main(int argc, char* argv[]) {
    int rank, size, data;

    MPI_Init(&argc, &argv);
```



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```
MPI_Comm_rank(MPI_COMM_WORLD, &rank);
MPI_Comm_size(MPI_COMM_WORLD, &size);

if (rank == 0) {
    data = 42; // Root sets data
}

MPI_Bcast(&data, 1, MPI_INT, 0, MPI_COMM_WORLD);
printf("Process %d received data %d\n", rank, data);

MPI_Finalize();
return 0;
}

Reduce Example (Summing Values)
#include <stdio.h>
#include <mpi.h>

int main(int argc, char* argv[]) {
    int rank, size, local_data, total_data;

    MPI_Init(&argc, &argv);
    MPI_Comm_rank(MPI_COMM_WORLD, &rank);
    MPI_Comm_size(MPI_COMM_WORLD, &size);

    local_data = rank + 1; // Each process has some local data

    MPI_Reduce(&local_data, &total_data, 1, MPI_INT,
MPI_SUM, 0, MPI_COMM_WORLD);

    if (rank == 0) {
        printf("Total sum of all data is: %d\n", total_data);
    }

    MPI_Finalize();
    return 0;
}
```



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Output:

```
lab1@localhost:~  
[lab1@localhost ~]$ nano hello_mpi.c  
[lab1@localhost ~]$ mpicc hello_mpi.c -o hello_mpi  
bash: mpicc: command not found...  
[lab1@localhost ~]$
```

```
lab1@localhost:~/openmpi-5.0.5  
are/doc/openmpi/{} \; ; \  
fi  
make[4]: Leaving directory '/home/lab1/openmpi-5.0.5/docs'  
make[3]: Leaving directory '/home/lab1/openmpi-5.0.5/docs'  
make[2]: Leaving directory '/home/lab1/openmpi-5.0.5/docs'  
make[1]: Leaving directory '/home/lab1/openmpi-5.0.5/docs'  
make[1]: Entering directory '/home/lab1/openmpi-5.0.5'  
make[2]: Entering directory '/home/lab1/openmpi-5.0.5'  
make install-exec-hook  
make[3]: Entering directory '/home/lab1/openmpi-5.0.5'  
make[3]: Leaving directory '/home/lab1/openmpi-5.0.5'  
make[2]: Nothing to be done for 'install-data-am'.  
make[2]: Leaving directory '/home/lab1/openmpi-5.0.5'  
make[1]: Leaving directory '/home/lab1/openmpi-5.0.5'  
[lab1@localhost openmpi-5.0.5]$ export PATH=$HOME/.openmpi/bin:$PATH  
export LD_LIBRARY_PATH=$HOME/.openmpi/lib:$LD_LIBRARY_PATH  
[lab1@localhost openmpi-5.0.5]$ mpicc --version  
gcc (GCC) 11.5.0 20240719 (Red Hat 11.5.0-5)  
Copyright (C) 2021 Free Software Foundation, Inc.  
This is free software; see the source for copying conditions. There is NO  
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.  
[lab1@localhost openmpi-5.0.5]$ gcc -fopenmp -o hello_omp hello_omp.c
```

```
lab1@localhost:~ -- nano hello_mpi.c  
GNU nano 5.6.1 hello_mpi.c Modified  
MPI_Comm_size(MPI_COMM_WORLD, &size);  
  
// Print hello message from each process  
printf("Hello from process %d of %d\n", rank, size);  
  
// Finalize MPI  
MPI_Finalize();  
  
return 0;  
}
```

```
[lab1@localhost openmpi-5.0.5]$ nano hello_mpi.c  
[lab1@localhost openmpi-5.0.5]$ mpicc hello_mpi.c -o hello_mpi  
[lab1@localhost openmpi-5.0.5]$  
[lab1@localhost openmpi-5.0.5]$ mpicc hello_mpi.c -o hello_mpi  
[lab1@localhost openmpi-5.0.5]$ mpirun -np 4 ./hello_mpi  
Hello from process 1 of 4  
Hello from process 3 of 4  
Hello from process 0 of 4  
Hello from process 2 of 4  
[lab1@localhost openmpi-5.0.5]$
```



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```
GNU nano 5.6.1 send_recv.c Modified
#include <stdio.h>
#include <mpi.h>

int main(int argc, char* argv[]) {
    int rank, size, data;

    MPI_Init(&argc, &argv);
    MPI_Comm_rank(MPI_COMM_WORLD, &rank);
    MPI_Comm_size(MPI_COMM_WORLD, &size);

    if (rank == 0) {
        data = 100;
        MPI_Send(&data, 1, MPI_INT, 1, 0, MPI_COMM_WORLD);
        printf("Process 0 sent data %d to process 1\n", data);
    }
    else if (rank == 1) {
        MPI_Recv(&data, 1, MPI_INT, 0, 0, MPI_COMM_WORLD, MPI_STATUS_IGNORE);
        printf("Process 1 received data %d from process 0\n", data);
    }
}
```

```
[lab1@localhost openmpi-5.0.5]$ nano send_recv.c
[lab1@localhost openmpi-5.0.5]$ mpicc send_recv.c -o send_recv
mpirun -np 2 ./send_recv
```

```
Process 0 sent data 100 to process 1
Process 1 received data 100 from process 0
[lab1@localhost openmpi-5.0.5]$ nano broadcast.c
```

```
GNU nano 5.6.1 broadcast.c Modified
#include <stdio.h>;
#include <mpi.h>;

int main(int argc, char* argv[]) {
    int rank, size, data;
    MPI_Init(&argc, &argv);
    MPI_Comm_rank(MPI_COMM_WORLD, &rank);
    MPI_Comm_size(MPI_COMM_WORLD, &size);
    if (rank == 0) {
        data = 42; // Root process sends data
    }
    // Broadcast data from process 0 to all other processes
    MPI_Bcast(&data, 1, MPI_INT, 0, MPI_COMM_WORLD);
    printf("Process %d received data %d\n", rank, data);
    MPI_Finalize();
    return 0;
}
```

```
[lab1@localhost openmpi-5.0.5]$ nano broadcast.c
[lab1@localhost openmpi-5.0.5]$ mpicc broadcast.c -o broadcast
mpirun -np 4 ./broadcast
```



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```
lab1@localhost:~/openmpi-5.0.5— nano broadcast.c
GNU nano 5.6.1 broadcast.c Modified
#include <stdio.h>
#include <mpi.h>

int main(int argc, char* argv[]) {
    int rank, size, data;

    MPI_Init(&argc, &argv);
    MPI_Comm_rank(MPI_COMM_WORLD, &rank);
    MPI_Comm_size(MPI_COMM_WORLD, &size);

    if (rank == 0) {
        data = 42; // root process sets the data
        printf("Process 0 is broadcasting data %d\n", data);
    }

    // Broadcast data from process 0 to all processes
    MPI_Bcast(&data, 1, MPI_INT, 0, MPI_COMM_WORLD);

    printf("Process %d received data %d\n", rank, data);
}
```

```
[lab1@localhost openmpi-5.0.5]$ nano broadcast.c
[lab1@localhost openmpi-5.0.5]$ mpicc broadcast.c -o broadcast
[lab1@localhost openmpi-5.0.5]$ mpirun -np 4 ./broadcast
Process 0 is broadcasting data 42
Process 0 received data 42
Process 2 received data 42
Process 1 received data 42
Process 3 received data 42
[lab1@localhost openmpi-5.0.5]$
```

```
[lab1@localhost openmpi-5.0.5]$ nano reduce.c
```

```
lab1@localhost:~/openmpi-5.0.5— nano reduce.c
GNU nano 5.6.1 reduce.c Modified
#include <stdio.h>
#include <mpi.h>

int main(int argc, char* argv[]) {
    int rank, size;
    int value, sum;

    MPI_Init(&argc, &argv);
    MPI_Comm_rank(MPI_COMM_WORLD, &rank);
    MPI_Comm_size(MPI_COMM_WORLD, &size);

    // Each process sets a value equal to its rank
    value = rank;

    // Reduce operation: sum of all values, result stored in root (process 0)
    MPI_Reduce(&value, &sum, 1, MPI_INT, MPI_SUM, 0, MPI_COMM_WORLD);

    if (rank == 0) {
        printf("Sum of all ranks = %d\n", sum);
    }
}
```

```
[lab1@localhost openmpi-5.0.5]$ nano reduce.c
[lab1@localhost openmpi-5.0.5]$ mpicc reduce.c -o reduce
Process 3 received data 42
[lab1@localhost openmpi-5.0.5]$ nano reduce.c
[lab1@localhost openmpi-5.0.5]$ mpicc reduce.c -o reduce
[lab1@localhost openmpi-5.0.5]$ mpirun -np 4 ./reduce
```

```
[lab1@localhost openmpi-5.0.5]$ nano reduce.c
[lab1@localhost openmpi-5.0.5]$ mpicc reduce.c -o
[lab1@localhost openmpi-5.0.5]$ mpirun -np 4 ./red
Sum of all ranks = 6
[lab1@localhost openmpi-5.0.5]$
```



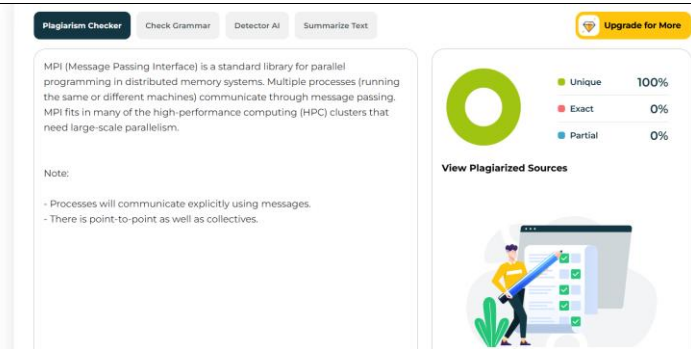

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Output Analysis	Together, these programs cover basic MPI features: ● Process identification (Hello World) ● Point-to-point communication (Send/Recv) ● Collective communication (Broadcast, Reduce)
Link of student Github profile where lab assignment has been uploaded	
Conclusion	MPI Programming implemented successfully.
Plag Report (Similarity index < 12%)	
Date	2 / 9 /25